

Project Memory Log: Morphological Processing ERP Paper

Conversation Summary and Continuity Document

1. The Original Problem

The user is writing a specialist psycholinguistics journal paper examining individual differences in morphological processing using ERPs and reaction times. The study involves a lexical decision task (LDT) with:

- **Participants:** 67 adults (60 with ERP data)
- **Stimuli:** 100 words and 100 morphologically complex nonwords
- **Nonword types:** Complex (real stem + real affix, e.g., *gasful*) vs. Simple (real stem + pseudoaffix, e.g., *gasfil*)
- **Measures:** Reaction times (RTs) and ERPs (N250 and N400 components)
- **Individual differences:** Two orthogonal PCA dimensions derived from vocabulary knowledge, spelling ability, and print exposure:
 - **Dim.1 = Language Proficiency (LP)** — depth/breadth of lexical-semantic knowledge
 - **Dim.2 = Orthographic Sensitivity (OS)** — sensitivity to sublexical orthographic structure

The user was struggling with: (a) theoretical framing of the introduction, (b) analytic strategy, (c) R code for mixed effects models, and (d) writing up results.

2. Key Theoretical Decisions

The Core Prediction: Double Dissociation

The study tests whether OS and LP dissociably modulate morphological processing at distinct temporal stages:

- **OS → early form-based processing (N250):** OS should modulate stem/base frequency effects at the N250, reflecting sensitivity to sublexical orthographic structure.
- **LP → later semantic integration (N400):** LP should modulate family size effects at the N400, reflecting depth of lexical-semantic knowledge.

This prediction follows from Andrews and colleagues' finding that orthographic and semantic profiles are dissociable axes of individual variation, combined with the theoretical distinction between stem frequency (form-level access) and family size (semantic representation richness).

The Three-Question Analytic Structure

The analyses are organised around three core questions:

1. **Q1 (RT/Behavioral):** Does OS influence overall lexical decision efficiency?
2. **Q2 (N250/Early neural):** Does OS modulate early morpho-orthographic parsing?
3. **Q3 (N400/Late neural):** Does LP shape semantic integration of morphological information?

Key Theoretical Models Discussed

- **Baayen–Schreuder race model** (classical dual-route): stem frequency → decomposition route; family size → whole-form route
- **Early automatic decomposition (Rastle & Davis):** obligatory early form-based parse, then semantic filter
- **Morpho-orthographic account (Grainger):** intermediate morpho-orthographic level
- **NDL/LDL (Baayen et al.):** no morpheme representations; discriminative learning from distributional statistics; both effects emergent from learned weight structure

3. Introduction — Current Status: Complete

Structure

1. Opening paragraphs on reading and English morphology
2. Bridge paragraph planting individual differences logic
3. **Subsection: Stem Frequency and Morphological Family Size: Theoretical Interpretations Across Current Models** — covers classical, early decomposition, morpho-orthographic, and NDL/LDL accounts
4. **Subsection: Individual Differences in Morphological Processing** — Lexical Quality Hypothesis; Andrews et al. (2013) finding; OS/LP profiles mapped onto theoretical accounts
5. **Subsection: Aims and Predictions** — three analyses, double dissociation as central claim

Key Framing Decisions

- Words and nonwords framed as doing distinct theoretical work (words = time course of stored representations; nonwords = generalisation to novel forms)
- RT models framed as efficiency measures (no morphological $\times$ ID interactions by design — stated explicitly in methods)
- ERP models framed as architecture measures
- The asymmetric interaction structure (OS at N250, LP at N400) is justified in both the aims section and the results section

4. Analytic Strategy — Finalised

RT Models

No morphological structure $\times$ individual difference interactions in RT models by design. Individual differences entered as main effects only.

Words RT:

```
m_rt_words <- lmer(
  logRT ~ zLogBF + zLogFS + OS + LP +
    (1 | SubjID) + (1 | Item),
  data = rt_w,
  control = lmerControl(optimizer = "bobyqa")
)
```

Nonwords RT:

```
m_rt_nw <- lmer(
  logRT ~ complexity + zLogBF + zLogFS + OS + LP +
    (1 + complexity | SubjID) + (1 | ItemID),
  data = rt_nw,
  control = lmerControl(optimizer = "bobyqa")
)
```

Contrast Coding

- `complexity`: sum coded $c(-0.5, 0.5)$ with levels $c("Simple", "Complex")$
- `family_size`: sum coded $c(-0.5, 0.5)$ with levels $c("Small", "Large")$
- `base_frequency`: sum coded $c(-0.5, 0.5)$ with levels $c("Low", "High")$
- OS and LP: PCA scores, already mean-centred by construction — no additional scaling needed

ERP Models — Base Frequency Analyses (Primary for Words)

Words N250 (Base Frequency $\times$ OS):

```
contrasts(erp_w$base_frequency) <- c(-0.5, 0.5)
m_w_N250_bf <- lmer(
  value ~ base_frequency * OS + LP +
    (1 | SubjID) + (1 | SubjID:chlabel),
  data = subset(erp_w, time_window == "N250"),
  control = lmerControl(optimizer = "bobyqa")
)
```

Words N400 (Base Frequency $\times$ LP):

```
m_w_N400_bf <- lmer(
  value ~ base_frequency * LP + OS +
    (1 | SubjID) + (1 | SubjID:chlabel),
  data = subset(erp_w, time_window == "N400"),
  control = lmerControl(optimizer = "bobyqa")
)
```

ERP Models — Family Size Analyses

Words N250 (Family Size $\times$ OS) — secondary:

```
contrasts(erp_w$family_size) <- c(-0.5, 0.5)
m_w_N250_fs <- lmer(
  value ~ family_size * OS + LP +
    (1 | SubjID) + (1 | SubjID:chlabel),
  data = subset(erp_w, time_window == "N250"),
  control = lmerControl(optimizer = "bobyqa")
)
```

Words N400 (Family Size $\times$ LP) — secondary:

```
m_w_N400_fs <- lmer(
  value ~ family_size * LP + OS +
    (1 | SubjID) + (1 | SubjID:chlabel),
  data = subset(erp_w, time_window == "N400"),
  control = lmerControl(optimizer = "bobyqa")
)
```

Nonwords N250 (Complexity $\times$ OS and LP; family_size as covariate — confirmed final):

```
contrasts(erp_nw$family_size) <- c(-0.5, 0.5)
m_nw_N250 <- lmer(
  value ~ complexity * OS + complexity * LP + family_size +
    (1 | SubjID) + (1 | SubjID:chlabel),
  data = subset(erp_nw, time_window == "N250"),
  control = lmerControl(optimizer = "bobyqa")
)
```

Nonwords N400 (Complexity $\times$ Family Size $\times$ LP — centerpiece finding):

```
contrasts(erp_nw$complexity) <- c(-0.5, 0.5)
contrasts(erp_nw$family_size) <- c(-0.5, 0.5)
m_nw_N400 <- lmer(
  value ~ complexity * family_size * LP + OS +
```

```
(1 | SubjID) + (1 | SubjID:chlabe1),
data = subset(erp_nw, time_window == "N400"),
control = lmerControl(optimizer = "bobyqa")
)
```

R Packages Used

```
library(lme4)
library(lmerTest)      # Satterthwaite df -- load BEFORE lmerTest,
                        BEFORE fitting models
library(emmeans)       # follow-up contrasts; use emm_options(lmer.df =
                        "kenward-roger")
library(effectsize)    # eta_squared(model, partial = TRUE)
library(performance)  # r2_nakagawa(model) -- use this, NOT r2()
library(MuMIn)        # r.squaredGLMM() as alternative if performance
                        fails
library(ggplot2)
library(patchwork)     # load after ggplot2, before lmerTest
```

Important Technical Notes

- Always load `ggplot2` and `patchwork` before `lmerTest` to avoid namespace conflicts
- Use `r2_nakagawa()` not `r2()` — the latter has a bug with certain model structures
- Set `emm_options(lmer.df = "kenward-roger")` at start of `emmeans` session
- For back-transforming log RT: use `tran = "log"` in `emmeans` or exponentiate manually
- KR df hang with complex random effects — Satterthwaite is acceptable and standard
- Nonword RT: `emmeans` crashed until `emm_options(lmer.df = "satterthwaite")` was set globally

5. Key Results — Summary

RT Results

Words RT:

- Base Frequency: $\beta = -0.017$, $t(100.97) = -3.66$, $p < .001$, $\eta_p^2 = .12$
- Family Size: $\beta = -0.013$, $t(93.59) = -2.64$, $p = .010$, $\eta_p^2 = .07$
- OS: marginal trend, $\beta = -0.035$, $t(63.84) = -1.85$, $p = .069$ — NOT robust to random effects changes
- LP: null, $p = .35$
- $R_{\text{marginal}}^2 = .033$, $R_{\text{conditional}}^2 = .417$

Nonwords RT:

- Complexity: $\beta = 0.049$, $t(62.68) = 9.13$, $p < .001$, $\eta_p^2 = .57$; $M_{\text{complex}} = 718.9$ ms [695.2, 743.4], $M_{\text{simple}} = 684.6$ ms [661.5, 708.5]; difference ≈ 34 ms
- Base Frequency: inhibitory effect, $\beta = 0.015$, $t(98.21) = 3.05$, $p = .003$, $\eta_p^2 = .09$
- Family Size: ns, $p = .22$
- OS: significant, $\beta = -0.041$, $t(62.21) = -2.23$, $p = .030$, $\eta_p^2 = .07$
- LP: null, $p = .78$
- $R_{\text{marginal}}^2 = .050$, $R_{\text{conditional}}^2 = .494$

N250 Results

Words N250 — Base Frequency (primary, tests double dissociation):

- Base Frequency main effect: $\beta = -0.318$, $t(1618) = -3.18$, $p = .002$; High more negative than Low
- **Base Frequency $\times$ OS:** $\beta = -0.455$, $t(1618) = -4.06$, $p < .001$ [*Key finding*]
- Follow-ups: BF effect absent at low OS ($p = .374$), significant at mean OS ($\Delta = 0.318$, $p = .002$), large at high OS ($\Delta = 0.773$, $p < .001$)
- LP: not interacted; main effect ns ($p = .162$)
- $R^2_{\text{marginal}} = .018$, $R^2_{\text{conditional}} = .801$

Words N250 — Family Size (secondary):

- Family Size: $\beta = -0.324$, $t(1618) = -5.54$, $p < .001$; $M_{\text{large}} = -0.57 \mu\text{V}$ $[-1.03, -0.11]$, $M_{\text{small}} = -0.90 \mu\text{V}$ $[-1.36, -0.44]$
- Family Size $\times$ OS: marginal trend, $\beta = -0.111$, $t(1618) = -1.70$, $p = .089$
- LP: ns
- $R^2_{\text{marginal}} = .019$, $R^2_{\text{conditional}} = .738$

Nonwords N250:

- Complexity: $\beta = -0.233$, $t(4856) = -4.63$, $p < .001$; $M_{\text{complex}} = -0.759 \mu\text{V}$, $M_{\text{simple}} = -0.526 \mu\text{V}$
- Family Size (covariate): $\beta = 0.182$, $t(4856) = 3.63$, $p < .001$
- **Complexity $\times$ OS:** $\beta = 0.392$, $t(4856) = 6.98$, $p < .001$ [*Key finding*]
- **Complexity $\times$ LP:** $\beta = 0.358$, $t(4856) = 8.76$, $p < .001$ [*Key finding*]
- Both interactions show same pattern: large complexity cost at low skill $\rightarrow$ attenuated $\rightarrow$ reversed at high skill:
 - OS: $\Delta = 0.642$ (low), 0.250 (mean), -0.142 (high, $p = .053$)
 - LP: $\Delta = 0.574$ (low), 0.216 (mean), -0.141 (high, $p = .032$)
- Theoretical interpretation: N250 reflects processing cost not sensitivity; reversal = real suffix facilitates parsing for skilled readers; both dimensions converge on early processing (qualifies predicted OS-specific effect)

N400 Results

Words N400 — Base Frequency (primary, tests double dissociation):

- Base Frequency main effect: $\beta = 0.214$, $t(1618) = 2.01$, $p = .044$; High more positive than Low (facilitation)
- **Base Frequency $\times$ LP:** $\beta = 0.344$, $t(1618) = 3.98$, $p < .001$ [*Key finding*]
- Follow-ups: BF effect absent at low LP ($p = .333$), significant at mean LP ($\Delta = -0.214$, $p = .044$), large at high LP ($\Delta = -0.558$, $p < .001$)
- OS: main effect significant $p = .049$ (overall amplitude, no interaction with BF)
- $R^2_{\text{marginal}} = .026$, $R^2_{\text{conditional}} = .825$

Words N400 — Family Size (secondary):

- Family Size: $\beta = -0.452$, $t(1618) = -7.07$, $p < .001$; $M_{\text{large}} = 0.518 \mu\text{V}$ $[-0.054, 1.089]$, $M_{\text{small}} = 0.062 \mu\text{V}$ $[-0.509, 0.634]$
- Family Size $\times$ LP: ns, $p = .172$
- OS: main effect significant, $p = .038$

Nonwords N400:

- Complexity: $\beta = -0.460$, $t(4854) = -7.48$, $p < .001$
- **Complexity $\times$ Family Size $\times$ LP three-way:** $\beta = 0.227$, $t(4854) = 2.27$, $p = .023$ [*Centerpiece finding*]
- Complexity $\times$ Family Size interaction by LP level:
 - Low LP: $\Delta = -0.667$, $p < .001$ (large, significant)
 - Mean LP: $\Delta = -0.440$, $p = .0003$ (attenuated, significant)
 - High LP: $\Delta = -0.213$, $p = .188$ (absent)

- Family size effect within complex nonwords disappears at high LP
- Complexity $\times$ Base Frequency $\times$ LP three-way: NOT significant (confirms specificity — LP effect is about family richness, not stem familiarity)
- Complexity $\times$ LP (averaged over FS): monotonic attenuation (unlike N250 reversal), significant at all LP levels

The Double Dissociation — Confirmed

	N250 (early form-based)	N400 (later semantic)
Base Frequency $\times$ OS	✓ Significant	OS main effect only
Base Frequency $\times$ LP	LP not interacted	✓ Significant

For nonwords: both OS and LP modulate N250 (partial qualification of predicted dissociation); LP selectively drives N400 three-way.

6. Results Section — Current Status

Written and Approved Paragraphs

Words section:

- ✓ Words RT — complete, verified against R output
- ✓ Words N250 (family size version) — complete
- ✓ Words N400 (family size version) — complete
- ✓ Words N250 (base frequency version) — drafted in final exchange
- ✓ Words N400 (base frequency version) — drafted in final exchange

Nonwords section:

- ✓ Nonwords RT — complete, verified against R output
- ✓ Nonwords N250 — complete with follow-up contrasts
- ✓ Nonwords N400 — complete with three-way follow-up contrasts

Summary section:

- ✓ Results Summary — drafted and approved, split so that the “three key findings” paragraph moves to open the Discussion

Results Section Structure

```
\section{Results}
  [Orienting paragraph]
  [Table 1: Model summary]
  [Asymmetric interaction explanation paragraph]

\subsection{Words}
  \paragraph{Words RT}
  \paragraph{Words N250}    % <- base frequency version now primary
  \paragraph{Words N400}    % <- base frequency version now primary

\subsection{Nonwords}
  \paragraph{Nonwords RT}
  \paragraph{Nonwords N250}
  \paragraph{Nonwords N400}

\subsection{Results Summary}
```

Key Writing Decisions

- Words and nonwords reported separately (not by component)
- RT then ERP (N250 then N400) within each stimulus type
- No claim that “interactions were absent in RTs” since they were not tested — instead state models were specified for efficiency only
- ERP measures framed as “designed to detect” modulation that RT models were not designed to detect
- N250 direction for base frequency: high frequency = MORE negative (opposite to family size effect) — to be explained in Discussion
- Conditional R^2 values for ERP models are very high (0.738–0.825) due to large participant and electrode variance

7. Outstanding Issues to Resolve

Critical (resolve before opening new chat)

1. **Nonword N250 covariate** — **RESOLVED:** Final model uses `family_size` as covariate.
2. **Integration of base frequency and family size analyses for words:** The words section now has two sets of ERP analyses — base frequency (primary, tests double dissociation) and family size (secondary). Need to decide:
 - Report base frequency as primary, family size as secondary/supplementary?
 - Or report both in full within the words section?
 - *Recommended:* base frequency primary in main text, family size results noted briefly or in supplementary.
3. **N250 direction for base frequency:** High frequency bases → MORE negative N250 (opposite of family size effect). This dissociation between BF and FS at N250 needs a sentence in Discussion. Likely explanation: BF = strength of form-level activation (more effort); FS = efficiency of pattern matching (less effort).

Important (address in new chat)

4. **Discussion** — **not yet started.** Structure agreed:
 - Opening: three key findings paragraph (moved from Results Summary)
 - Section 1: N250 reversal at high skill (most unexpected finding)
 - Section 2: Behavioral/ERP dissociation
 - Section 3: N400 three-way and semantic integration (connect to theoretical frameworks)
 - Closing: theoretical implications and limitations
5. **Updated model summary table:** Table 1 needs updating to reflect base frequency as primary predictor for words ERP analyses (currently shows family size).
6. **Results orienting paragraph:** Needs minor update to reflect that base frequency analyses are primary for words.
7. **Nonword N400 base frequency analyses:** Null three-way (Complexity × BF × LP) should be reported as a cross-check confirming specificity of the family size finding — currently mentioned but not formally integrated into the results write-up.

8. Discussion — Planned Structure (not yet written)

Opening Paragraph (moved from Results Summary)

The three key findings that structure the Discussion:

1. Individual differences dissociate from behavioral responses but are visible in neural dynamics
2. LP selectively drives the N400 three-way (closest support for predicted dissociation)
3. N250 shows reversal at high skill; N400 shows attenuation without reversal — qualitatively different mechanisms

Section 1: Early Morphological Parsing and the N250 Reversal

- Processing cost vs. morphological sensitivity interpretation
- Why the reversal indicates real suffix facilitates parsing for skilled readers
- Why both OS and LP converge on early processing — broad lexical experience, not just orthographic skill
- Marginal Family Size $\times$ OS for words — consistency with nonword pattern
- Connect to morpho-orthographic and discriminative learning accounts

Section 2: The Behavioral/ERP Dissociation

- Individual differences invisible in RTs but visible in ERPs
- Implications for measuring individual differences in morphological processing
- Lexical Quality Hypothesis — precision of representation vs. speed of response
- Why OS predicts nonword RT but not morphological modulation in ERPs

Section 3: N400 Three-Way and Semantic Integration

- What Complexity $\times$ Family Size $\times$ LP means
- Attenuation without reversal — contrast with N250
- LP selective at N400; OS not — closest to predicted dissociation
- Connect to NDL/LDL: LP indexes depth of form-to-meaning mapping
- Connect to multi-stage accounts
- Null Family Size $\times$ LP for words — why LP does not modulate FS for familiar words
- Base Frequency $\times$ LP at N400 for words — LP enhances semantic integration benefit of high frequency stems

Closing: Theoretical Implications and Limitations

- Partial dissociation — what it means
- Limitations: unmatched words, asymptotic CIs, generalisability
- Future directions

9. Key Terminology and Variable Names

Concept	Variable name in R	Notes
Language Proficiency	LP, Dim.1	PCA dimension 1, already centred
Orthographic Sensitivity	OS, Dim.2	PCA dimension 2, already centred
Base/Stem Frequency	zLogBF (RT), base_frequency (ERP)	Continuous for RT, binned for ERP
Family Size	zLogFS (RT), family_size (ERP)	Continuous for RT, binned for ERP
Complexity	complexity	Factor: Simple vs. Complex
Participant ID	SubjID	
Item ID	Item (words), ItemID (nonwords)	Note different names
ERP amplitude	value	
Electrode nested in participant	SubjID:chlabel	
Log RT	logRT	log(response_time)

10. Citations Needing Completion

The following citations appear as [CITATION] placeholders throughout the manuscript:

- Baayen & Schreuder race model — original dual-route paper
- Rastle, Davis et al. — early automatic decomposition / masked priming papers
- Grainger et al. — morpho-orthographic account

- Baayen et al. NDL paper — 2011 *Psychological Review*
- Baayen et al. LDL paper — ~2019 *Complexity* journal
- Family size effects — original Schreuder & Baayen paper
- Stem frequency operationalisation — relevant citations
- Semantic transparency and family size — opacity studies
- N250 and N400 components — relevant ERP citations
- Andrews et al. (2013) — individual differences in morphological priming
- Perfetti — Lexical Quality Hypothesis (2002, 2007, 2017)
- Andrews and colleagues — lexical precision papers
- Marslen-Wilson, Tyler, Devlin — neuroimaging morphology papers

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